

UNITED INTERNATIONAL UNIVERSITY																																																															
Department of Computer Science and Engineering (CSE)																																																															
Course Syllabus																																																															
1.	Course Title	Data Structure and Algorithms I																																																													
2.	Course Code	CSE 2215																																																													
3.	Trimester and Year	Summer 2023																																																													
4.	Pre-requisites	CSE 1111 Structured Programming Language																																																													
5.	Credit Hours	3.00																																																													
6.	Class Hours	<table border="1"> <thead> <tr> <th>SECTION</th> <th>DAY</th> <th>TIME</th> <th>ROOM #</th> </tr> </thead> <tbody> <tr> <td>E</td> <td>Sat & Tue</td> <td>8.30 – 9:51</td> <td>304</td> </tr> </tbody> </table>											SECTION	DAY	TIME	ROOM #	E	Sat & Tue	8.30 – 9:51	304																																											
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7.	Counselling Hours	Will be uploaded to lms																																																													
8.	Instructor's Name	Md Shadman Aadeeb																																																													
9.	Email	shadman@cse.uiu.ac.bd																																																													
10.	Office	Room # 0719 (D)																																																													
11.	Text Book	Data Structures - E M Reingold																																																													
12.	Reference	- Introduction to Algorithms-Thomas H Cormen - Fundamentals of Computer Algorithm-Sartaj Sahni																																																													
13.	Course Contents	Internal data representation; Abstract data types; Introduction to algorithms; Asymptotic analysis: growth of functions, O, Ω and Θ notations; Correctness proof and techniques for analysis of algorithms; Master Theorem; Elementary data structures: arrays, linked lists, stacks, queues, trees and tree traversals, graphs and graph representations, heaps, binary search trees; Graph Traversals: DFS, BFS, Applications of DFS and BFS; Sorting: heap sort, merge sort, quick sort, linear-time sorting; Data structures for set operations.																																																													
14.	Course Outcomes (COs)	<table border="1"> <thead> <tr> <th>COs</th> <th>Description</th> </tr> </thead> <tbody> <tr> <td>CO1</td> <td>Able to choose appropriate data structure as applied to specified problem definition.</td> </tr> <tr> <td>CO2</td> <td>Able to handle operations like searching, insertion, deletion, traversing mechanism etc. on various data structures.</td> </tr> <tr> <td>CO3</td> <td>Able to use linear and non-linear data structures like stacks, queues, linked list etc.</td> </tr> </tbody> </table>											COs	Description	CO1	Able to choose appropriate data structure as applied to specified problem definition.	CO2	Able to handle operations like searching, insertion, deletion, traversing mechanism etc. on various data structures.	CO3	Able to use linear and non-linear data structures like stacks, queues, linked list etc.																																											
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18. Lecture Outline

Class	Topics/Assignments	CO Mapping
1.	Introduction to Data Structure: linear data structure Complexity Analysis, Big Oh, Big Omega	
2.	Lower and Upper Limit, Best and Worst Cases	
3.	Arrays: Memory Mapping, Linear and Binary Search, Linear Time Sorting (Counting Sort)	CO2, CO3
4.	Arrays: Sorting, Merge Sort	CO1
5.	Arrays: Sorting, Quick Sort	CO1
6.	Linked Lists: Single Linked List	CO2, CO3
7.	Linked Lists: Double Linked List, Circular Linked List	CO2, CO3
8.	Linked Lists: Application	CO1
9.	Stack: Implementation using Array and Linked Lists	CO2, CO3
10.	Stack: Application, Tower of Hanoi, Recursion	CO1
11.	Infix to Postfix and Postfix evaluation, Introduction to Queue	CO1, CO2, CO3
12.	Implementation of Queue using Arrays and Linked lists	CO1, CO2, CO3
MID TERM EXAMINATION WEEK		
13.	General Tree, Binary Tree Implementation using Array and left-right pointers	CO2, CO3
14.	Tree Traversal Algorithms, Tree Applications	CO2, CO3
15.	Binary Search Trees, Insertion, Deletion	CO2, CO3
16.	Binary Search Tree Properties	CO1, CO2, CO3
17.	Heap and Priority Queue	CO1
18.	Heap Sort and Application of Priority Queue	CO1, CO2, CO3
19.	Graphs: Implementation using adjacency matrix and Adjacency lists Sparse and Dense graph, Space requirements	CO2, CO3
20.	Graphs: BFS using adjacency matrix and Adjacency lists	CO2, CO3
21.	Graphs: DFS using adjacency matrix and Adjacency lists	CO2, CO3
22.	Graphs: Application of Graphs, Directed Acyclic Graph, Topological Sort	CO1, CO2, CO3
23.	Set Operations	CO1, CO2
24.	Set Operations Continued	CO1, CO2
FINAL EXAMINATION		

Appendix 1: Assessment Methods

CO	Assessment Method	(%)
-	Attendance	5
-	Assignment	5
-	Class Tests	20
CO1, CO2, CO3, CO4	Midterm Exam	30
CO1, CO2, CO3	Final Exam	40

Appendix 2: Grading Policy

Letter Grade	Marks %	Grade Point	Letter Grade	Marks%	Grade Point
A (Plain)	90-100	4.00	C+ (Plus)	70-73	2.33
A- (Minus)	86-89	3.67	C (Plain)	66-69	2.00
B+ (Plus)	82-85	3.33	C- (Minus)	62-65	1.67
B (Plain)	78-81	3.00	D+ (Plus)	58-61	1.33
B- (Minus)	74-77	2.67	D (Plain)	55-57	1.00
			F (Fail)	<55	0.00

Appendix-3: Program outcomes

POs	Program Outcomes
PO1	An ability to apply knowledge of mathematics, science, and engineering
PO2	An ability to identify, formulate, and solve engineering problems
PO3	An ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability
PO4	An ability to design and conduct experiments, as well as to analyze and interpret data
PO5	An ability to use the techniques, skills, and modern engineering tools necessary for engineering practice
PO6	The broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context
PO7	A knowledge of contemporary issues
PO8	An understanding of professional and ethical responsibility
PO9	An ability to function on multidisciplinary teams
PO10	An ability to communicate effectively
PO11	Project Management and Finance
PO12	A recognition of the need for, and an ability to engage in life-long learning